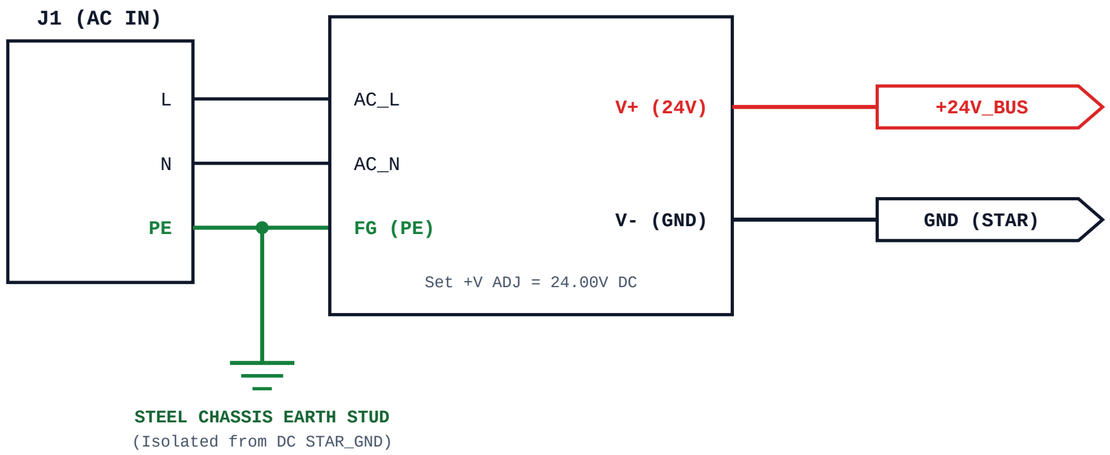
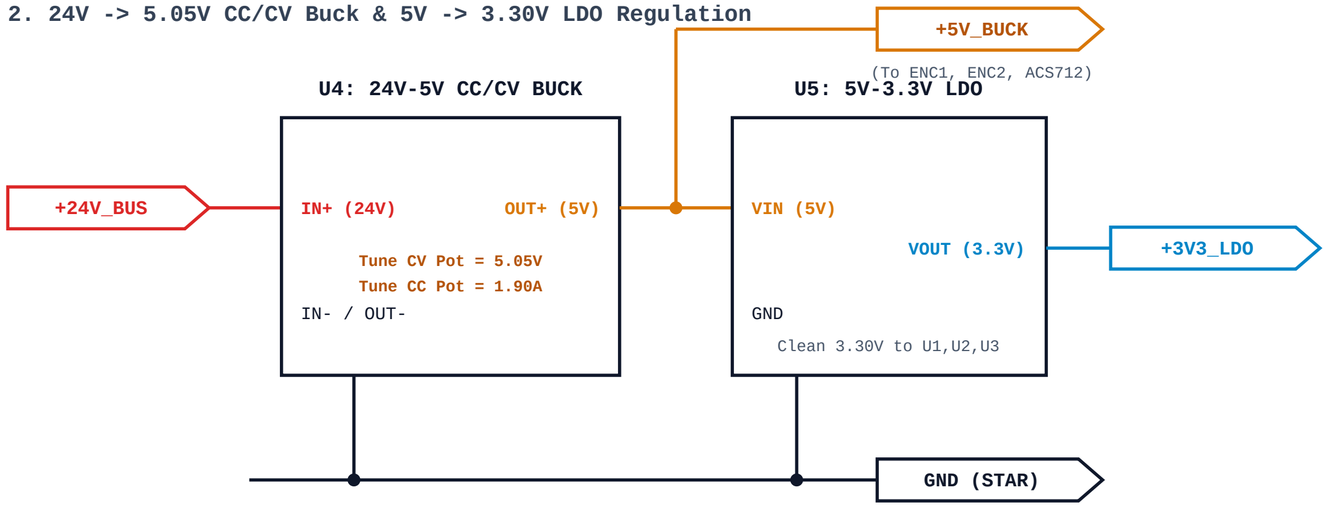
Sheet 1: Power Supply, CC/CV Buck-LDO Regulation, Steel Chassis Earth & Master ESP32-S

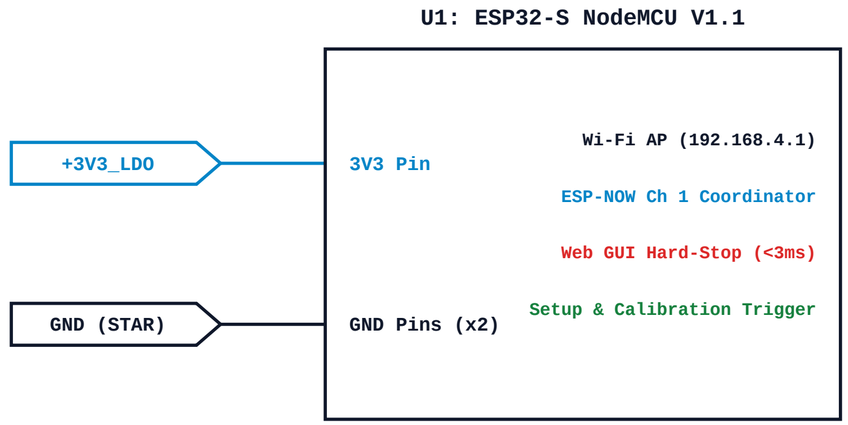
1. 24V 30A SMPS Power Supply & Steel Chassis Earthing



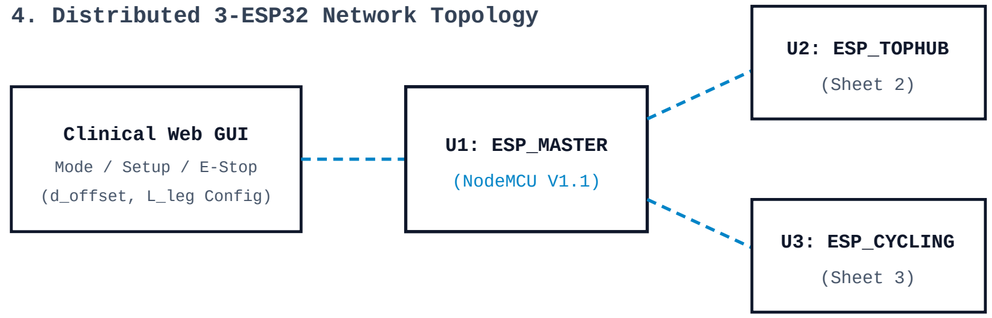
2. 24V -> 5.05V CC/CV Buck & 5V -> 3.30V LDO Regulation



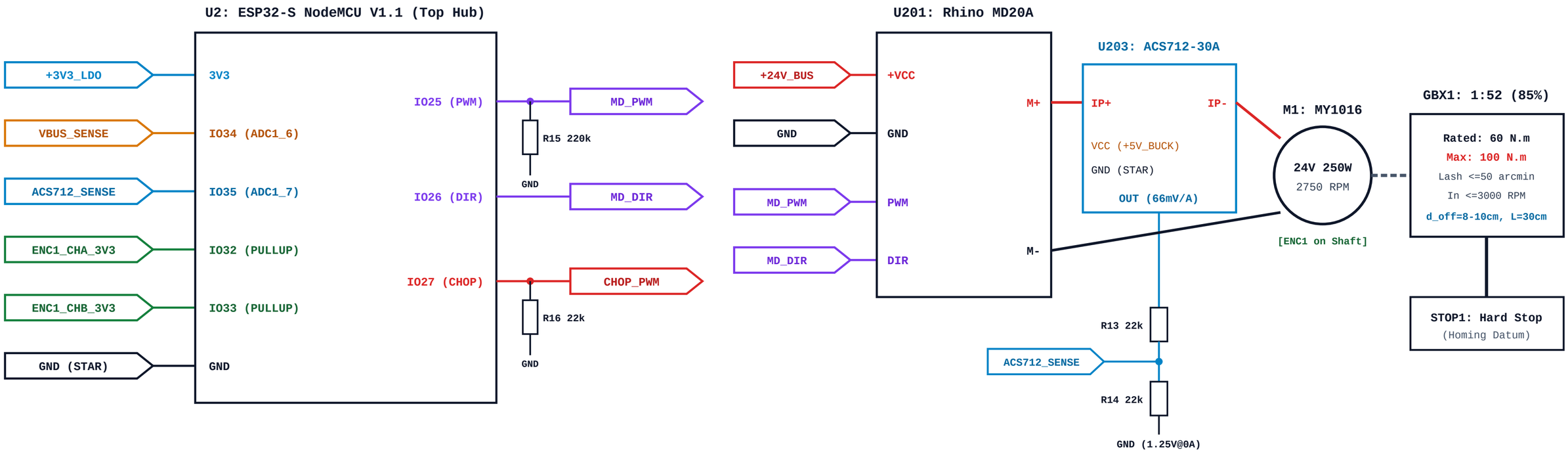
3. Master Node U1 (ESP32-S NodeMCU V1.1)



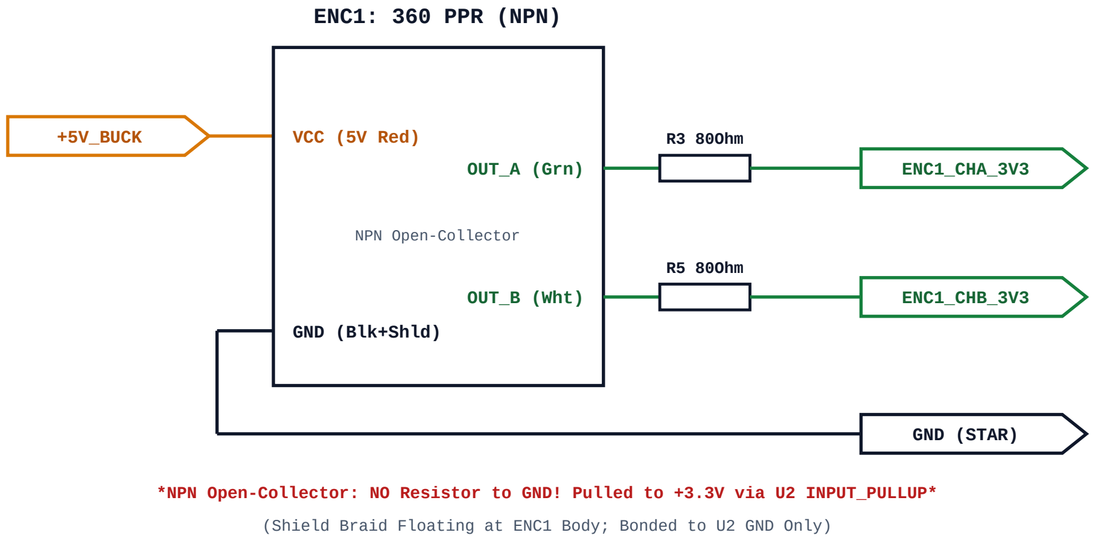
4. Distributed 3-ESP32 Network Topology



Sheet 2: Top Hub Node (U2), ACS712-30A, 85% Gearbox, Series Chopper & NPN Encoder



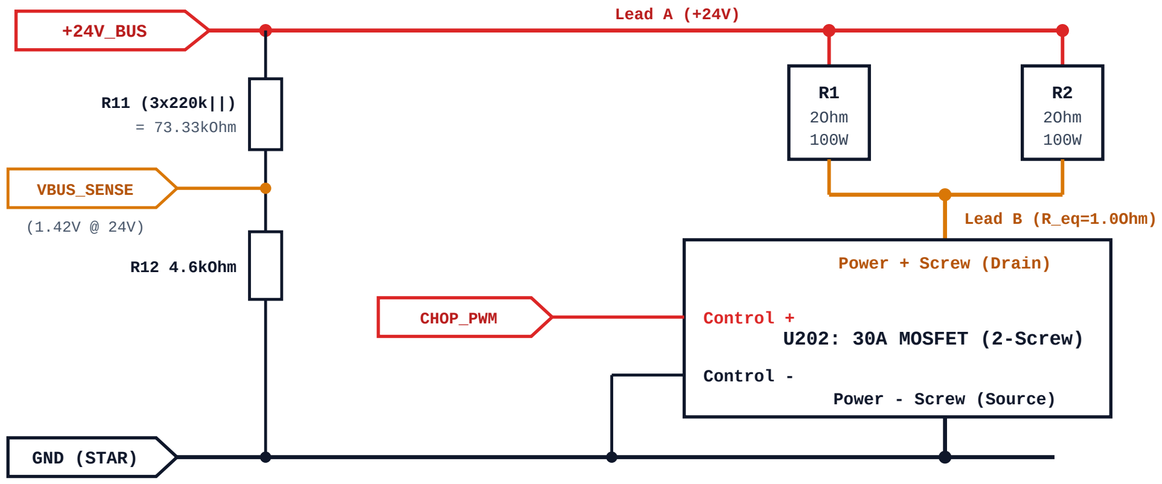
3. Top-Hub Motor-Shaft Optical Encoder (NPN Open-Collector, 74,880 CPR)



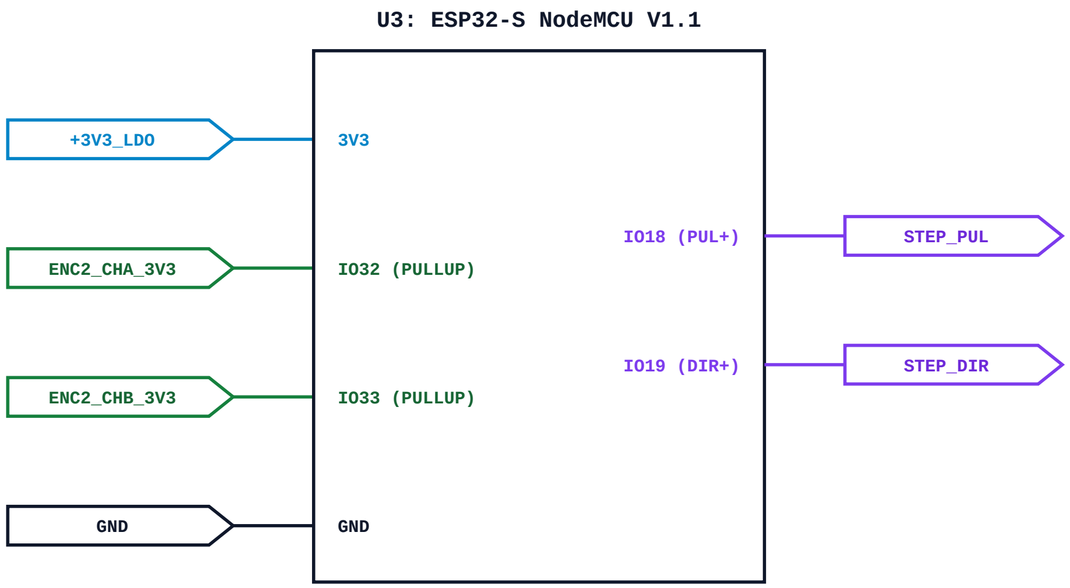
NPN Open-Collector: NO Resistor to GND! Pulled to +3.3V via U2 INPUT_PULLUP

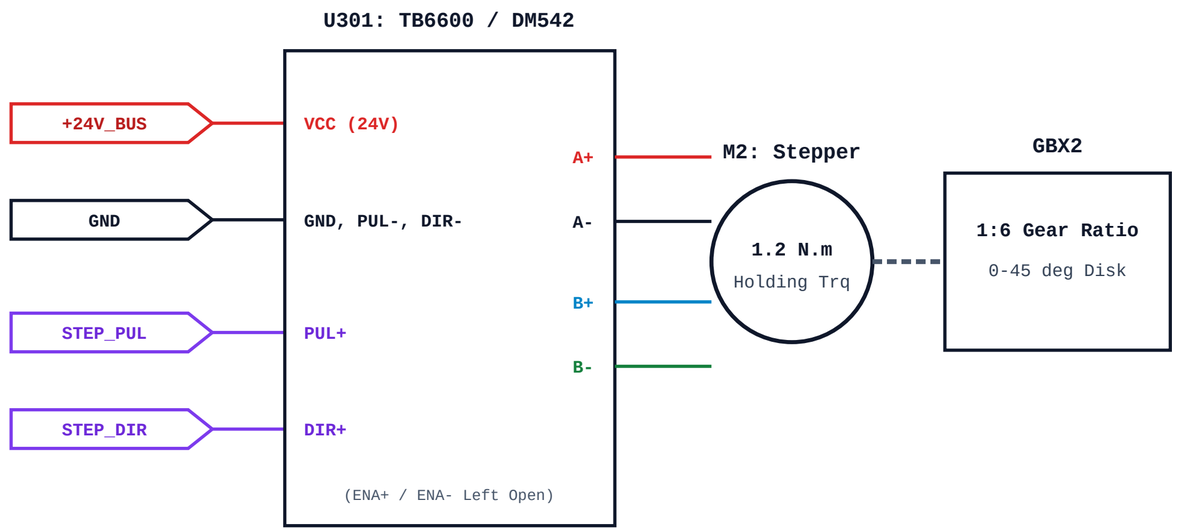
(Shield Braid Floating at ENC1 Body; Bonded to U2 GND Only)

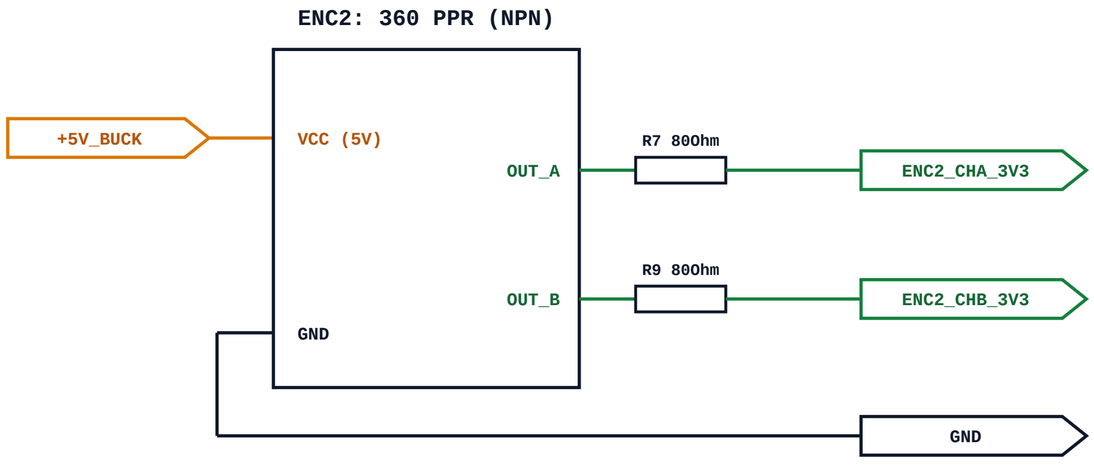
4. VBUS Divider (3x220k|| / 4.6k) & Series Brake Chopper (R1||R2 -> U202)



Sheet 3: Bottom Cycling Eddy Current Brake, Stepper & Flywheel (NPN Open-Collector ENC2)

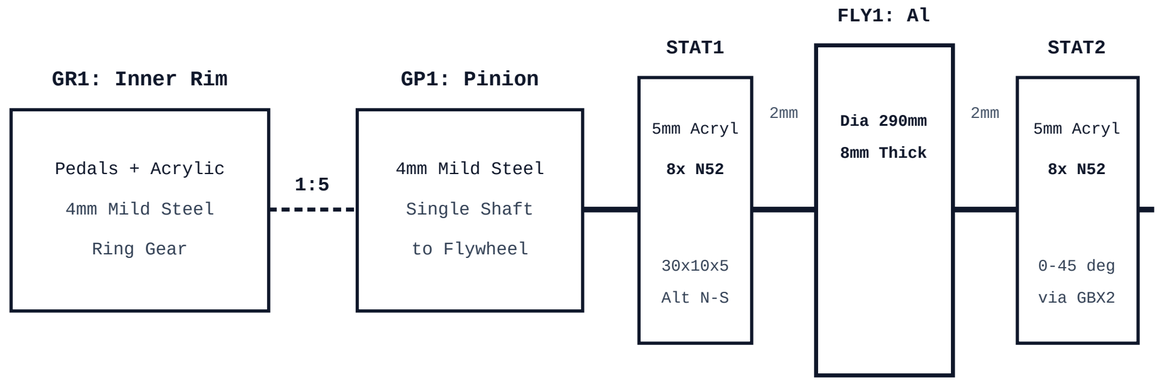






NPN Open-Collector: NO Resistor to GND! Pulled to +3.3V via U3 INPUT_PULLUP

(Shield Braid Floating at ENC2 Body; Bonded to U3 GND Only)



Sheet 4: Bill of Materials (BOM) & Verified Resistor Inventory Allocation

REF DES	COMPONENT	QTY	SPECIFICATION / VERIFIED WIRING DESCRIPTION
PSU1	24V 30A SMPS Power Supply	1	24V DC, 30A (720W) Switching PSU; PE/FG bonded to Steel Chassis (Sheet 1)
U4	24V-to-5V CC/CV Buck	1	Tuned to CV = 5.05V, CC = 1.90A for Encoders, ACS712 & LDO Input (Sheet 1)
U5	5V-to-3.3V LDO Regulator	1	Tuned to 3.30V for U1, U2, U3 ESP32-S NodeMCU 3V3 Pin (Sheet 1)
U1, U2, U3	ESP32-S NodeMCU V1.1	3	U1: Master Web GUI, U2: Top Knee Hub, U3: Bottom Cycling (50Hz ESP-NOW, <3ms E-Stop)
U201	Rhino MD20Amp Motor Driver	1	6V-30V, 20A Single-Channel DC Driver; 20kHz PWM on IO25, DIR on IO26 (Sheet 2)
U203	ACS712-30A Current Sensor	1	+/-30A Hall Sensor (66 mV/A, 5V VCC) inline between U201 M+ and M1 (Sheet 2)
M1	E-Bike DC Motor MY1016	1	24V DC, 250W, 2750 RPM PMDC Motor; Case Isolated from DC M+/M- (Sheet 2)
GBX1, STOP1	1:52 Gearbox & Hard Stop	1	1:52 ratio, 85% eff, 60 N.m rated / 100 N.m peak, <=50 arcmin backlash (Sheet 2)
U202	30A Heatbed MOSFET (2-Pin)	1	2-Screw Low-Side Switch: Power+ from R1||R2, Power- to STAR_GND (Sheet 2)
R1, R2	2.0 Ohm 100W Brake Bank	2	Aluminum-shell resistors in parallel (1.0 Ohm, 200W) between +24V & U202 Power+
ENC1, ENC2	360 PPR NPN Encoders	2	NPN Open-Collector (5V VCC); ENC1 on M1 shaft (74,880 cnt/rev), ENC2 on Flywheel
R3,R5,R7,R9	80 Ohm Damping Resistors	4	4x 80 Ohm series resistors on ENC1 & ENC2 OUT_A/B (+ ESP32 INPUT_PULLUP; NO GND R)
R11, R12	24V Bus Sense Divider	4	R11 = 3x 220kOhm parallel (73.33kOhm) + R12 = 1x 4.6kOhm (1.417V @ 24V to IO34)
R13, R14	ACS712 Sense Divider	2	R13 = 1x 22kOhm + R14 = 1x 22kOhm (1:1 divider, 1.250V @ 0A, 33 mV/A to IO35)
R15, R16	PWM Pull-Down Resistors	2	R15 = 1x 220kOhm (IO25 to GND), R16 = 1x 22kOhm (IO27 to GND) [Spares: 4x117R, 330R]
U301,M2,GBX2	1.2 N.m Stepper & TB6600	1	1.2 N.m stepper + TB6600/DM542 (PUL-/DIR- to GND, ENA open) + 1:6 gear GBX2 (Sheet 3)
GR1,GP1,FLY1	1:5 Gear Set & Al Flywheel	1	4mm steel 1:5 ring/pinion + 290x8mm Al Flywheel + SKF 6205-Z, 6004-2RSL, 619/6-2Z
STAT1, STAT2	Symmetrical N52 Stators	2	5mm acrylic sheets, 2mm air gap, 8x (30x10x5mm) N52 magnets/side (0-45 deg overlap)

Resistor Inventory Summary: 4x 220kOhm (3 in R11, 1 in R15) | 3x 22kOhm (R13, R14, R16) | 1x 4.6kOhm (R12) | 4x 80Ohm (R3, R5, R7, R9) | Spares: 4x 117Ohm, 1x 330Ohm